

Benchmark Report

Benchmarking of NQCH's quantum computer

August 8, 2026

1. Report of Changes

Platform: sinq20

Calibration-id: b91dd3f15f1127994cdcf515616216e8cb9cbb56

Calibration date: 2026-08-04 09:10:26

Calibration note: chore(sinq20): update sq params

Experiment-id: 20260808212202

Experiment date: 2026-08-08 21:22:03

Experiment note: temporary note!!!

Platform: sinq20

Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78

Calibration date: 2026-01-15 02:03:03

Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538

Experiment date: 2026-01-18 16:15:38

Experiment note: temporary note!!!

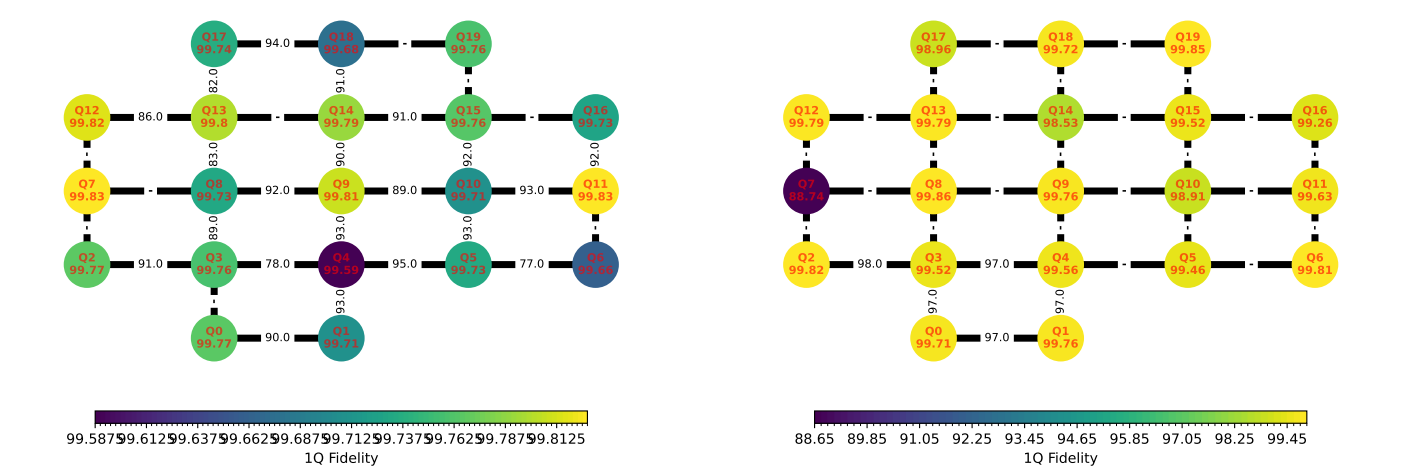
2. Version Comparison

Library	Version	Library	Version
qibo	0.3.3	numpy	2.5.1
qibolab	0.2.17	qibocal	0.2.6
matplotlib	3.11.1	scipy	1.18.0
scikit-learn	1.9.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.13.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



4. Statistics

	Average	Median	Min	Max
T1 (ns)	24.8	25.1	14.1	31.7
T2 (ns)	11.2	11.6	3.76	21.8
Fidelity	0.983	0.99	0.949	0.998
RO Fidelity	0.981	0.981	0.976	0.988
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981
2q CZ Fidelity	0.975	0.974	0.967	0.991

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	4, 5	0.955
3	1, 4, 5	0.944
4	1, 4, 5, 10	0.939
5	1, 4, 5, 10, 11	0.938

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark	Metric	Value	Runtime
Mermin	Max	3.408	226.42 sec
Grover 2q	Max prob	0.902	2.86 sec
Grover 3q	Max prob	0.342	3.76 sec
GHZ state prep.	Success	0.922	3.00 sec

Benchmark	Metric	Value	Runtime
Mermin	Max	-3.316	1.02 sec
Grover 2q	Max prob	0.89	2.36 sec
Grover 3q	Max prob	0.385	2.43 sec
GHZ state prep.	Success	0.910	3.95 sec

7. Randomized Benchmarking Results

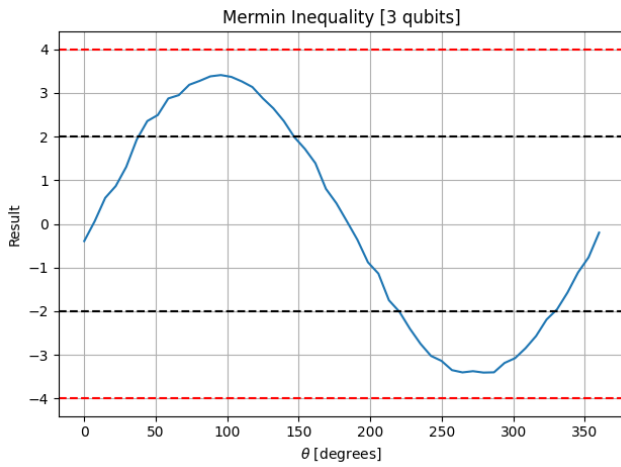
Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.998	± 9.3e-05	10	0.997	± 0.000162
1	0.997	± 9.4e-05	11	0.998	± 6.63e-05
2	0.998	± 7.29e-05	12	0.998	± 6.58e-05
3	0.998	± 9.66e-05	13	0.998	± 0.000102
4	0.996	± 0.000271	14	0.998	± 8.09e-05
5	0.997	± 0.000146	15	0.998	± 0.000103
6	0.997	± 0.000168	16	0.997	± 0.000162
7	0.998	± 6.72e-05	17	0.997	± 8.82e-05
8	0.997	± 0.000103	18	0.997	± 0.000138
9	0.998	± 0.000155	19	0.998	± 8.11e-05

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131	10	0.989	± 0.000582
1	0.998	± 7.47e-05	11	0.996	± 0.000218
2	0.998	± 7.57e-05	12	0.998	± 0.000126
3	0.995	± 0.000163	13	0.998	± 7.15e-05
4	0.996	± 0.000185	14	0.985	± 0.00183
5	0.995	± 0.000147	15	0.995	± 0.000469
6	0.998	± 8.34e-05	16	0.993	± 0.000558
7	0.887	± 0.0287	17	0.99	± 0.000502
8	0.999	± 4.48e-05	18	0.997	± 0.00026
9	0.998	± 7.2e-05	19	0.998	± 9.05e-05

8. Mermin

Mermin's algorithm for 3 qubits.

Runtime	Qubits	Max
226.42 sec	1, 4, 5	3.408



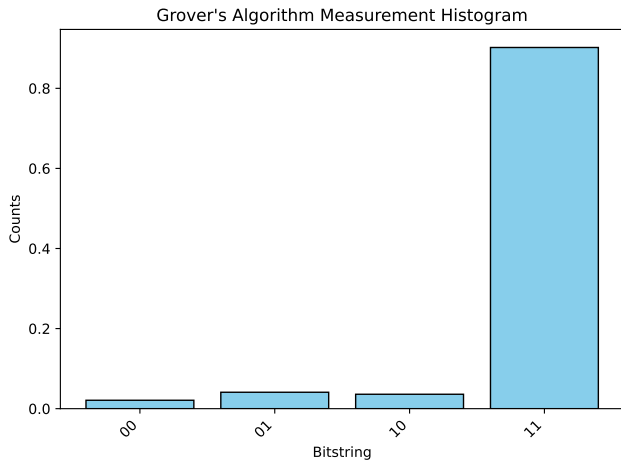
Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[4, 5]].

Runtime	Qubits	Max prob
2.86 sec	4, 5	0.902



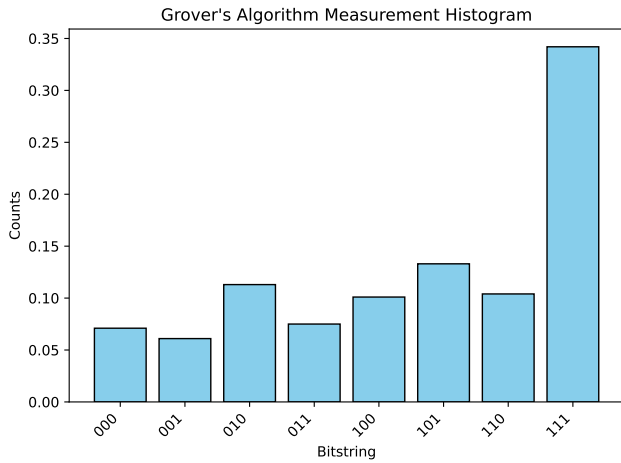
Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89



10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in [[1, 4], [4, 5]].

Runtime	Qubits	Max prob
3.76 sec	1, 5, 4	0.342



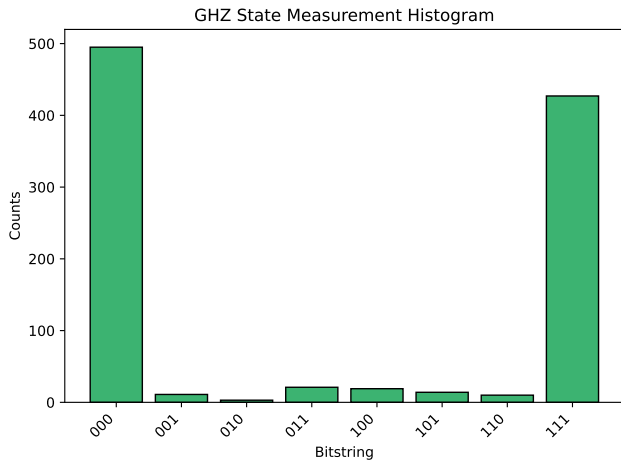
Runtime	Qubits	Max prob
2.43 sec	0, 2, 3	0.385



11. GHZ state preparation

GHZ circuit with 3 qubits executed on sinq20 backend with 1000 shots.

Runtime	Qubits	Success
3.00 sec	1, 4, 5	0.922



Runtime	Qubits	Success
3.95 sec	0, 3, 2	0.910

